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Student-Oriented Chain-of-Thought Optimization

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[ABSTRACT]: This thesis examines a practical difficulty in mathematical reasoning distillation. In my experiments, some teacher-generated solutions looked correct and readable, yet they still did not function well as supervision for a smaller student model. After going through these cases one at a time, I gradually realized that the issue was usually not with the whole rationale, but with one local part of it. Sometimes the problem looked small but was hard to ignore. A bridge step might be missing, a symbolic transformation might be squeezed into one line, or some intermediate relation might never be written out. Starting from such cases, I use a student-oriented local CoT revision framework with three roles: Teacher, Student, and Controller. The Teacher provides the original rationale and several candidate repairs for the local region. The Student helps show where the reasoning becomes hard to learn from at the step level. The Controller then uses these signals to decide which part is worth revising and how much revision is needed. The repaired traces are later used to fine-tune Qwen2.5-Math student models under a unified math500 strict evaluation setting. The results suggest a fairly clear point: changing a small but important local region can make the reasoning data more useful for training. But that does not mean an easier-to-learn rationale will automatically produce the same level of gain in standalone problem solving.

[Key words]: Chain-of-Thought; mathematical reasoning; student model learning; supervised fine-tuning; data optimization

[摘要]：本文讨论数学推理蒸馏里的一个实际问题：教师模型生成的思维链即使答案正确、整体上也基本通顺，直接拿来训练小参数学生模型时，却未必同样有效。我在实验中反复见到，问题通常不在整条推理链，而是落在某个局部位置，例如关键过渡被省掉、符号变换写得过密，或者中间量之间的关系没有交代清楚。对教师模型来说，这类压缩表达往往还能依靠自身能力补上；但对主要依赖模仿学习的学生模型而言，这些地方恰恰容易成为学习阻塞。基于这一观察，本文提出一个面向学生模型学习的局部思维链优化框架，由 Teacher、Student 和 Controller 三部分组成：Teacher 负责生成原始思维链及局部修复候选，Student 提供步骤层面的学习难度信号，Controller 再据此判断哪些局部位置更值得干预，并完成候选筛选。随后，本文基于 QLoRA+DoRA 对 Qwen2.5-Math 系列学生模型进行监督微调，并在统一的 math500 strict 协议下进行评测。实验结果表明，在不改动整体解题路径的前提下，对局部瓶颈步骤做有控制的修补，通常能够提升思维链作为训练数据的使用价值；但“更容易学”和“独立解题更强”并不能被简单视作同一件事。因此，在教师思维链进入学生模型训练时，正确性、表达清晰度与学生侧可学习性应当被分别看待。

[关键词]：思维链；数学推理；学生模型学习；监督微调；数据优化

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1 Introduction

1.1 Research Background and Significance

The problem studied in this thesis did not come from a purely abstract idea. It grew out of repeated difficulties I encountered during the actual process of constructing training data. While working with teacher-generated mathematical solutions, I noticed that many of them looked completely fine when read on their own. In most cases, they were correct, logically consistent, and sometimes even pleasantly concise. But once these same solutions were used as supervision for a smaller student model, the training gains were often much less stable than expected. After running into this kind of mismatch again and again, it no longer felt reasonable to treat it as random noise in the experiments.

Mathematical reasoning makes this kind of mismatch easier to notice than many other tasks. In a long derivation, the final answer may still be correct even though one local step has been written too briefly. Sometimes a substitution is skipped, sometimes two symbolic operations are folded into a single line, and sometimes an intermediate relation is never stated explicitly at all. For a strong teacher model, these missing links are often not a serious problem, because it can fill them in using its own prior knowledge. A smaller student model trained mainly through imitation does not have that same luxury. What it receives is only the written trace.

For this reason, answer-only supervision is often too coarse for reasoning tasks. On some problems, the final label already contains most of the useful information. Mathematical benchmarks such as MATH are different because successful solving usually depends on a multi-step path rather than a short endpoint [2]. CoT prompting made this issue much easier to observe, since intermediate reasoning is no longer hidden behind the final answer [1]. Once those intermediate steps are treated as training targets, the way they are written starts to matter in its own right.

There is also a straightforward practical reason to care about this issue. The strongest model in a pipeline is not always the model that can finally be deployed. Smaller student models remain attractive because they are cheaper to train, easier to serve, and more realistic in constrained settings. However, transferring reasoning from teacher to student is not automatic [7, 8]. A rationale may look natural to a human reader and still be a poor object for imitation. In this thesis, I study this gap between teacher-side adequacy and student-side learnability as the central problem.

1.2 Research Motivation and Problem Setting

What made this problem worth pursuing was a repeated sample-level observation. In many cases, the original teacher CoT was not globally wrong, so fully rewriting it would have been hard to justify. But once I examined the trace step by step, the same kind of pattern kept appearing: one short local region carried too much unstated information. After enough examples of this type, the more useful question was no longer whether the whole rationale looked good or bad. The more useful question was where the student was likely to get stuck.

This distinction becomes important under supervised fine-tuning. The student is trained token by token. If a bridge step is missing, then the compressed jump itself becomes part of the supervision. In that sense, the student is not only learning mathematical content. It is also learning the teacher's way of writing, including the habit of skipping links that the teacher can internally reconstruct. A stronger model may recover those links almost effortlessly. A smaller one may fail exactly at that point.

Because of this, I do not treat a CoT as a single indivisible block of text. I treat it as a chain of local steps that play different roles. Some steps are routine. Some hold the structure of the whole solution together. Some look short on the page but are still difficult for a small model to absorb. This way of looking at the problem suggests a more restrained intervention strategy. Rather than rewriting the whole rationale from beginning to end, it makes more sense to first identify the small region that is most likely to cause difficulty for the student, and then revise only that part.

The experimental setting here is kept deliberately close to the way the project works in reality. The Teacher is treated mainly as a black-box model for generation and local revision, rather than something whose internal states can be directly used. The Student is a smaller Qwen2.5-Math model, and it plays two roles at the same time. On the one hand, it is the model we ultimately want to train. On the other hand, it also helps indicate where the written reasoning becomes hard to absorb. The Controller uses these signals to decide what kind of local edit should actually be made. This setup is close to the real conditions of the project, where a stronger teacher model may only be accessible through an API, while the student model remains the part that can still be examined and fine-tuned in a more direct way.

1.3 Research Objectives and Thesis Framework

Based on the observations above, this thesis concentrates on four questions:

1. How can feedback from a student model be used to locate the part of a teacher-generated reasoning chain that is hardest to learn from?

2. How should local repair be designed so that the revised CoT becomes easier for the student without breaking the original solution path?

3. Under one fixed training and evaluation pipeline, how should original CoTs and repaired CoTs be compared as supervision data?

4. Can local repair bring measurable gains on mathematical benchmarks after downstream fine-tuning?

To handle these questions, I use a Teacher–Student–Controller framework in this thesis. The Teacher first produces the base rationale, and after that it is also used to generate several candidates for local repair. The Student is not added as a replacement for the Teacher. Instead, its role is to reveal which part of the written reasoning seems hardest for the student side to absorb. The Controller then takes these signals as a basis and decides whether a local intervention is needed, and if so, where it should be made. The idea behind this design is not to rewrite everything more aggressively than necessary. As long as the original reasoning path is still workable, it should be kept. But if a certain local step is under-explained and clearly makes imitation harder, then a small revision there is usually a better choice.

This work has both methodological and empirical value, though in a fairly specific sense. Methodologically, it offers a more concrete way to look at reasoning data below the level of the full sample. Empirically, it keeps the downstream setup fixed and compares original CoTs with locally repaired ones. What matters here is not whether every repaired trace can be called better in some general sense. The real question is narrower: whether targeted local revision can actually increase the training value of teacher-generated reasoning data for student models.

1.4 Thesis Organization

The rest of the thesis is structured in a fairly straightforward way. Chapter 2 looks at the prior work most relevant to this problem. Chapter 3 introduces the method proposed in this thesis, especially the parts on key-step localization, local repair, and candidate selection. Chapter 4 explains the datasets, student models, training setup, and evaluation protocol used in the experiments. Chapter 5 presents the main results and discusses what can reasonably be taken from them. The thesis then ends in Chapter 6 with a brief conclusion and several directions that may be worth exploring later.

It is helpful to clarify one point at the outset. In the following chapters, the training objective and the evaluation protocol are kept as stable as possible. This is done so that the contrast between original and repaired CoTs can be read mainly as a data-side effect, rather than being mixed together with several unrelated changes in the pipeline.

2 Related Work

2.1 Chain-of-Thought and Mathematical Reasoning

Research on Chain-of-Thought prompting is the most direct starting point for this thesis. Once large language models began to write out intermediate reasoning, evaluation no longer stopped at the last line alone; the route toward the answer also became visible [1]. That change matters here because the reasoning route is not used only for inference. In this work, it is later turned into supervision data.

The relevance is especially clear in mathematics. A mathematical solution is usually a chain of substitutions, transformations, and intermediate conclusions rather than a short answer string. The MATH benchmark was proposed precisely to evaluate this kind of multi-step problem solving [2]. In this setting, getting the final answer right does not automatically mean that the rationale is easy to learn from. A derivation can still be a poor training example if the key transition is rushed over or folded into one overly compact sentence.

This is also one place where the present thesis takes a somewhat different view from standard CoT discussions. Earlier work explains why intermediate reasoning deserves attention, but it does not directly answer a narrower transfer question: if a rationale helps a strong teacher solve a problem, does that automatically mean it is a good target for a smaller student model? In this thesis, the answer is treated as nontrivial. The two uses are related, but they are not the same.

A local view makes the difference easier to see. One omitted substitution or one compressed bridge may not stop the teacher from obtaining the correct answer, yet it can still change how learnable the full trace becomes for the student. This is one reason why the present work focuses on local repair rather than judging only complete trajectories.

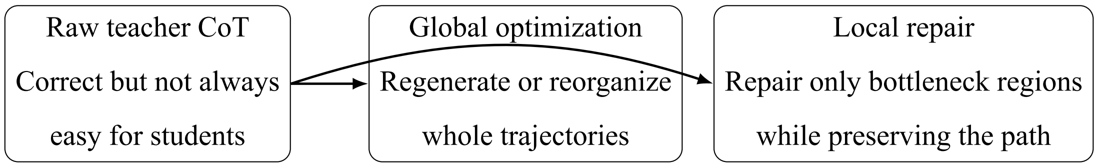

Figure 1 Two broad ways of improving reasoning traces. Existing data-centric methods often allow full-trajectory optimization, whereas this thesis focuses on local repair and tries to preserve the original solution path as much as possible.

2.2 Process Supervision and Step-level Analysis

A second line of work asks how reasoning should be examined once intermediate steps are visible. Final-answer correctness still matters, but it is no longer the only concern when the reasoning trace itself later becomes part of training data. A solution may finish with the right answer and still contain omitted justification, weak transitions, or symbolic relations that remain too implicit. Process-supervision research is useful in this context because it shifts attention away from the endpoint alone and toward the internal path [16].

Step-level analysis pushes this view further. Some studies argue that failure often appears not as one isolated wrong sentence, but as a break between neighboring steps [19]. Other work discusses recoverability, local contribution, or finer-grained supervision for smaller models [18, 15].These papers do not use the same pipeline as the one adopted in this thesis, but they do point to a common idea: structure below the level of the full sample can still matter in a meaningful way for learning.

What I take from this line of work is mainly a practical point. If the aim is to improve reasoning data for student models, then not every questionable step needs to be treated with the same level of concern. What matters more is the local region where difficulty and importance come together. This is also where the present thesis differs from evaluator-style process supervision. The purpose here is not mainly to check reasoning after it has already been written, but to identify which part of a teacher-generated CoT is actually worth revising before it is used for training.

Table 1 Representative Perspectives on Intermediate Reasoning in Prior Work

Research perspective	Main concern	Typical analysis unit	Relevance to this thesis
CoT prompting	Whether explicit reasoning helps inference	Full rationale / full trajectory	Shows that intermediate traces matter, especially on difficult reasoning tasks
Process supervision	Whether the reasoning process itself is reliable	Step, segment, or local transition	Suggests that internal weaknesses can be identified below the full-sample level
Step-flow analysis	Whether adjacent steps are properly connected	Transition between steps	Motivates focusing on missing bridges and abrupt local jumps
Step-wise learning	Whether small models need finer supervision	Step or action sequence	Supports the view that student models are sensitive to local reasoning structure

2.3 Reasoning Traces as Training Data

Another noticeable change in recent work is that reasoning traces are no longer viewed simply as fixed outputs. Instead of keeping or discarding an entire rationale as a whole, many methods now try to filter, combine, reconstruct, or revise these traces for downstream use [5, 6, 13, 14]. This change is closely connected to the main concern of the present thesis.

Compared with many trajectory-level approaches, the stance taken here is relatively conservative. In quite a few previous studies, regenerating the whole explanation is considered an acceptable choice.By contrast, the method studied in this thesis begins from a narrower assumption: many teacher traces are already mostly usable, and the main problem may sit in only one local region. If that is true, rewriting the whole rationale may be unnecessary, and in some cases it may even remove structure the student could still have learned from.

Mathematics is a good setting in which to look at this assumption more closely. A solution can still be globally correct even though one small but important gap remains inside the derivation. For the teacher model, that missing part may not look like a serious issue at all. But for the student model, it may turn out to be precisely the place where learning starts to break down. This observation is the most direct motivation for the local repair strategy adopted in this thesis.

2.4 Distillation, SFT, and Reasoning-oriented Post-training

The broader literature on distillation and reasoning-oriented post-training also provides part of the background for this thesis. One point that appears quite repeatedly in this line of work is that a strong teacher does not naturally or automatically turn into a strong student. Earlier studies suggest that transfer quality is shaped not just by the teacher itself, but also by the form of the reasoning traces, the level of granularity they use, and the way those traces are selected during transfer [7, 8, 9].

A similar point can also be seen from the training side. SRFT changes the downstream objective by combining supervised fine-tuning with reinforcement learning [3]. Light-R1 and AceReason likewise suggest that the final outcome can be influenced by a range of design choices, including data filtering, curriculum organization, and post-training strategy [10, 11].These methods are not direct baselines for the present thesis, but they still make one broader point quite clear: the quality of the data and the design of training usually matter together.

What makes this thesis different is not just the general idea that data quality matters. The real difference is where I choose to intervene. I am not mainly trying to redesign the optimization objective, and I am also not satisfied with only doing coarse sample-level filtering. What I want to examine instead is whether a teacher-generated CoT can be looked at from the student’ s perspective, whether the local part most likely to cause trouble can be found, and whether only that specific part can be revised before supervised training begins.

2.5 Summary and Positioning of This Thesis

Taken together, the literature above supports three ideas used throughout this thesis. First, intermediate reasoning is worth exposing rather than leaving hidden. Second, local properties of a reasoning chain can matter even when the final answer is correct. Third, reasoning traces can be revised instead of being treated as untouchable supervision. The present thesis builds on all three points, but keeps the scope intentionally narrow: it focuses on repairing the local bottleneck that matters most for a smaller student model.

This also makes the position of the thesis easier to state. It does not claim that local repair should replace every other form of reasoning-data optimization. It also does not claim that data-side editing alone can solve the transfer problem. The narrower question is more concrete: when a teacher trace is already correct, is it written in a form that a smaller model can genuinely learn from? The framework proposed here is meant as one direct way to examine that question.

3 Method

3.1 Overall Framework

The method used in this thesis follows a three-stage Teacher–Student–Controller framework. For each mathematical problem, the Teacher first generates an original CoT. The Student then reads the reasoning chain step by step and produces signals about where the trace seems easier or harder for a student model to learn from. The Controller uses these signals to judge which local part is actually worth changing, and the Teacher is then asked to make a repair in that region so that the final CoT becomes easier for student learning to absorb. Put more simply, the pipeline can be summarized as: mathematical problem data $\rightarrow$ Teacher generates the base CoT $\rightarrow$ Student analyzes the trace and gives feedback $\rightarrow$ key-step localization $\rightarrow$ local repair to obtain the final CoT $\rightarrow$ student fine-tuning $\rightarrow$ benchmark evaluation.

In this framework, the Teacher is responsible for giving the original reasoning and for producing candidates for local repair. The Student is not introduced as something that takes over the Teacher’s role as the solver. Its role is more diagnostic: it helps show which parts of the written reasoning seem difficult for a smaller model to take in. The Controller then uses these signals to decide what kind of data editing should actually be carried out.So the goal here is not to make yet another mathematical solver. What I really want to look at is whether existing teacher CoTs can be revised into a form that student models can pick up more easily.

To be more specific, the Teacher is treated as a *black-box* in this framework. I do not assume access to its hidden states, gradients, or internal reasoning activations. Instead, the method works with things that can actually be observed from outside the Teacher, including its outputs, the local repair candidates it generates, and the feedback signals computed on the Student side. In that sense, the Teacher is mainly used as an external provider of reasoning and local edits, while the signal that determines where intervention should be made comes from the Student–Controller side, not from any direct change inside the Teacher itself.

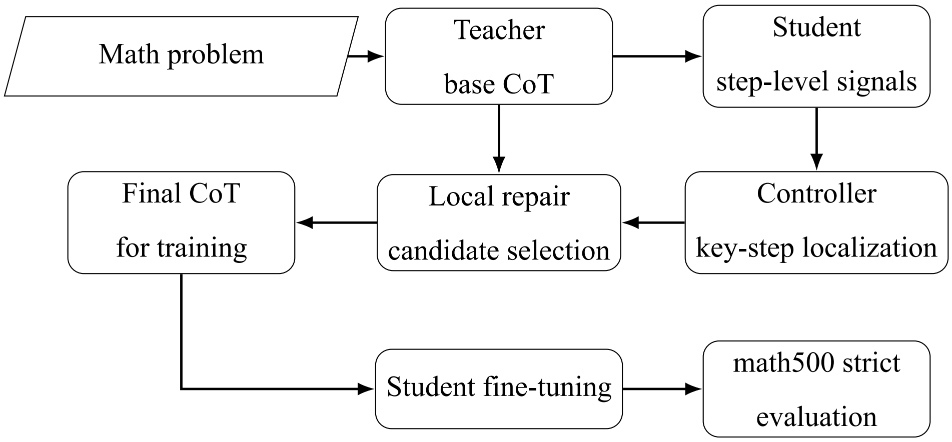

Figure 2 Overall workflow of the proposed Teacher–Student–Controller framework. The Teacher is responsible for producing the base CoT and local repair candidates, the Student indicates which parts seem harder to learn from at the step level, and the Controller uses these signals to choose the region for local revision before downstream fine-tuning and evaluation.

3.2 Problem Formulation

Let a mathematical problem be denoted by x, the Teacher-generated original reasoning chain by $z = (s_1, s_2, \ldots, s_n)$, where s_i is the i-th reasoning step, and the final answer by y. In conventional demonstration-based training, the triple (x, z, y) is typically used directly as a supervision example for the student model. In contrast, the method in this thesis edits only selected local steps in z while keeping the problem x and the main solution path largely unchanged, thereby producing an optimized reasoning chain $z' = (s'_1, s'_2, \ldots, s'_m)$.

Under this formulation, the method has to answer three questions: which step should be repaired, how should it be repaired, and which repaired candidate should finally be kept. The rest of this chapter explains these three parts in order.

3.3 Key-step Localization

In this method, a CoT is not treated as one whole piece of text that cannot be further examined. It is first divided into multiple steps, and the analysis then focuses on which positions are most likely to matter for student learning. The localization of key steps is based on three kinds of information taken together.

3.3.1 Outcome-level Signal

The first type is the outcome-level signal, which reflects how much a step affects the recovery of the final answer. A step may not look particularly difficult on the surface, but if it has a large influence on whether the student can recover the correct final answer, then it should not be ignored. In this thesis, such a signal reflects the importance of the step.

3.3.2 Process-level Difficulty Signal

The second type is the process-level difficulty signal, which describes how hard a step is for the student model to predict or absorb. The goal here is to answer questions such as whether this step seems substantially harder than nearby steps, whether it contains unusually dense information, whether it requires an extra bridge step, and whether it involves a large local reasoning jump. Compared with looking only at global answer loss, step-level difficulty signals are more suitable for locating internal bottlenecks in the CoT.

3.3.3 Structural Signal

The third type is the structural signal. What it tries to capture is whether a step carries dense information, key mathematical anchors, important variable relations, or major logical transitions. In mathematical reasoning, many important steps are difficult not only because of how they are written, but because several intermediate quantities, constraints, or symbolic transformations are often packed into the same part of the derivation. The role of structural signals is therefore to help distinguish between steps that merely look long and steps that are structurally central.

3.3.4 Integrated Scoring

To combine these three aspects, the method assigns each step a unified score $R_i = \alpha\widetilde{O}_i + \beta\widetilde{D}_i + \gamma\widetilde{S}_i$, where $\widetilde{O}_i$, $\widetilde{D}_i$, and $\widetilde{S}_i$ denote the normalized outcome-level, process-level, and structural signals for step s_i. In this thesis, the weights are set to $\alpha = 0.40$, $\beta = 0.40$, and $\gamma = 0.20$, with $\alpha + \beta + \gamma = 1$.

In practice, each signal is normalized within the current CoT by min–max scaling, namely $\widetilde{X}_i = \frac{X_i - \min_j X_j}{\max_j X_j - \min_j X_j + \epsilon}$, where ϵ is a small constant for numerical stability.This allows the three components to be compared on a shared scale before they are weighted.

These weights are chosen with the practical goal of the method in mind. The outcome-level signal and the process-level difficulty signal are given the same and relatively large weights because what the method mainly wants to find are steps that both matter for final-answer recovery and are genuinely hard for the student model to get through. The structural signal is still kept in the scoring process, but it plays a more supportive role and mainly helps make target-step selection more stable. Under this setting, the target step is selected as $i^\star = \arg\max_i R_i$.

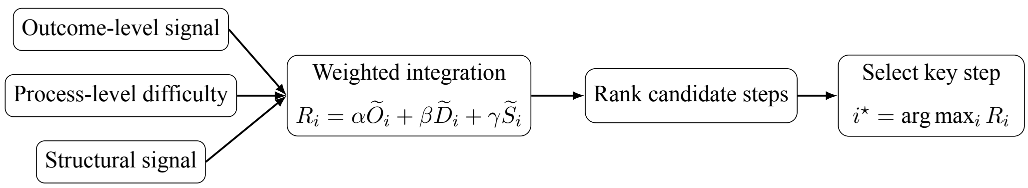

Figure 3 Integrated scoring for key-step localization. The method first normalizes the outcome-level, process-level, and structural signals, then combines them into a single score that is used to rank candidate steps and determine which region should be repaired.

3.4 Local Repair Strategies

Once the key region has been found, the method focuses on local repair rather than full-chain rewriting. Rewriting the whole reasoning chain would provide a larger editing space, but it would also raise the chance of damaging the original solution structure and bringing in extra noise that is not really needed. In many cases, the parts that make learning difficult for the student are concentrated in just a few local places. So compared with rewriting everything, small and targeted edits are usually the more stable choice.

To keep local repair more precise, this thesis does not treat every revision as the same kind of generic rewrite. Instead, several structured edit types are used. The main ones are listed below.

- **Pedagogical Rewrite**: rewrites a key step so that it becomes easier to follow and more instructional, while keeping the original solution structure largely unchanged.
- **Bridge Insertion**: introduces an intermediate transition step when the original connection between two steps is too abrupt.
- **Step Split**: divides a dense or wide-jump step into two or more finer-grained substeps.
- **Local Clarification / Correction**: makes small-scale clarifications or corrections to variables, symbols, quantity relations, or local expressions.
- **Answer Anchoring**: when necessary, adds a light answer anchor near the end of the

reasoning chain so that the final answer remains clearly recoverable after local edits.

These edits are not intended to make the reasoning uniformly longer. Their purpose is to make the most important regions of the reasoning chain easier for the student to absorb while preserving the original mathematical logic as much as possible.

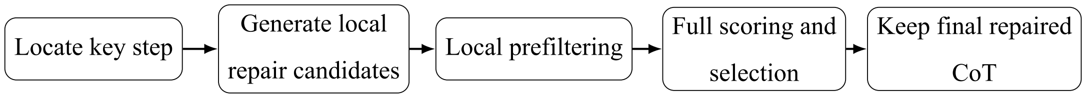

Figure 4 Selection pipeline after key-step localization. Once the bottleneck region has been located, the method generates several local repair candidates, removes the unstable ones through filtering, and finally keeps the highest-scoring repaired CoT for downstream training.

3.5 Candidate Generation and Multi-stage Selection

For each key region, the Teacher does not generate only one repair candidate. Several local candidates are produced so that the method does not end up depending on a single generation result, and so that there is still some room for later filtering. Rather than taking a candidate as soon as it appears, the method goes through a multi-stage selection process before deciding which one to keep.

3.5.1 Local Prefiltering

The first stage is lightweight local prefiltering, which removes clearly unsuitable candidates at low cost. Some candidates may be longer but fail to repair the actual missing bridge. Others may over-rewrite the original step and damage the local mathematical structure. The purpose of prefiltering is to eliminate such problematic candidates early, so that later full scoring can focus on more reasonable options.

In practice, candidates are filtered according to several local constraints, including whether the edit becomes too long, whether important mathematical anchors are still preserved, and whether the candidate changes the original structure too aggressively. Candidates that fail these basic checks are discarded before deeper evaluation. The set of candidates that pass this stage is denoted by $\mathcal{Z}_{\text{valid}}$, which is later used in the final selection step.

3.5.2 Full Scoring

Candidates that pass the prefilter then enter the full scoring stage. Full scoring evaluates four aspects simultaneously: answer recoverability, namely whether the repaired reasoning still supports stable recovery of the final answer; intermediate-step learnability, namely whether the repair truly reduces the difficulty of the hardest local region rather than only changing the wording; structural fidelity, namely whether important mathematical anchors, variable relations, and the main solution path are preserved; and edit magnitude control, namely whether the change becomes too large, too redundant, or too disruptive to the original intent of the solution.

The final candidate quality score is written as $Q(z^{(k)}) = \mu_1 A^{(k)} + \mu_2 G^{(k)} + \mu_3 F^{(k)} - \mu_4 C^{(k)}$, where $A^{(k)}$ represents answer recoverability, $G^{(k)}$ represents learnability gain, $F^{(k)}$ represents structural fidelity, and $C^{(k)}$ represents the edit cost of the k-th candidate. In practice, these quantities are implemented as normalized scores computed from final-answer recoverability, local difficulty reduction, structure preservation, and edit magnitude. So the formula is not introduced only for conceptual explanation. Each term is connected to a candidate property that can actually be measured and then used in the selection step.

In this thesis, the weights are set to $\mu_1 = 0.35$, $\mu_2 = 0.35$, $\mu_3 = 0.20$, and $\mu_4 = 0.10$. This setting is chosen to match the practical goal of the method. Answer recoverability and learnability gain receive the largest weights because the repaired CoT should remain useful for final answer recovery while also becoming easier for the student model to learn from. Structural fidelity is assigned a moderate weight, since preserving the mathematical logic of the original solution remains important. Edit cost receives the smallest weight, because overly large or disruptive edits should be penalized without dominating the final decision.

The final repaired CoT is then selected by $z^\star = \arg\max_{z^{(k)} \in \mathcal{Z}_{\text{valid}}} Q(z^{(k)})$, where $\mathcal{Z}_{\text{valid}}$ denotes the set of candidates that pass the prefiltering stage. If no candidate satisfies the required constraints, the original CoT is kept unchanged.

3.6 Summary of the Method Design

Overall, the method is built from three parts: key-step localization, structured local repair, and multi-stage candidate selection. What it tries to do is make the reasoning easier for the student model to learn from, without damaging answer recoverability or the core mathematical structure of the original solution.

4 Experimental Setup

4.1 Datasets and Models

The data used in this thesis come from a mathematical reasoning line related to SRFT. SRFT combines supervised fine-tuning and reinforcement learning, and its broader pipeline is built on OpenR1-Math-46k-8192 together with filtered high-quality reasoning responses [3]. I do not try to reproduce that entire recipe here. The scope of the present work is narrower. Starting from a fixed subset in the same broader data line, I ask a more focused question: if the wording of the rationale is changed while the problem itself stays the same, will a smaller student learn differently from it?

To make that comparison readable, I sample 5,000 mathematical problems and construct two paired supervision sets on this subset. For each problem, one version keeps the original teacher-generated CoT, and the other keeps a locally revised version obtained after bottleneck repair. This pairing matters because it removes an otherwise serious ambiguity. If both the problem set and the reasoning style changed at once, the later score difference would be much harder to interpret.

In the data construction stage, GPT-4o-mini is used as the Teacher. For each sampled problem, it first writes the base rationale, and after the target region is selected, it is also used to generate candidates for local repair. So in this thesis, the Teacher is mainly treated as a source of reasoning data and local editing, not as the model whose performance is being evaluated. The student models are Qwen2.5-Math-1.5B and Qwen2.5-Math-7B.

In this thesis, the downstream benchmark is fixed to math500. Of course, this does not imply that one benchmark alone is sufficient to represent the whole problem. The point is simply to keep the comparison concentrated. Across runs, the contrast remains the same: the same sampled problems, the same teacher family, and the same benchmark endpoint, but different forms of reasoning supervision.

4.2 Training Setup

After the data are prepared, the student models are trained separately on the original-CoT set and the repaired-CoT set. The training pipeline follows QLoRA+DoRA completion-style supervised fine-tuning rather than full-parameter SFT. This choice mainly reflects the practical constraints of the project, and it also helps keep the comparison controlled. The aim is to isolate the effect of changes in reasoning form instead of mixing that effect with a complete redesign of the optimization pipeline.

This also marks an important difference from SRFT. SRFT changes the training objective itself by combining supervised and reinforcement components [3]. The present work keeps the downstream optimization stage comparatively standard and instead changes the written reasoning that enters that stage. In that sense, the intervention here happens earlier and is more directly data-oriented.

Each training sample contains the mathematical problem, the target reasoning sequence, and the final answer form. Under this setup, the student is not learning only the last answer string. It is also learning how intermediate reasoning is written and organized. That is why local repair can matter. If the revised version explicitly writes out a transition that the original version compressed, then the supervision signal changes even though the underlying problem remains the same.

The best hyperparameters are not assumed to be identical for the 1.5B and 7B students. In actual experiments, the two model scales do not respond in exactly the same way to settings such as learning rate or gradient accumulation. So instead of forcing one training configuration onto both, this thesis reports the representative best results for each scale under the same evaluation protocol.

4.3 Evaluation Setup

All final results are reported under a unified math500 strict protocol. At test time, the student receives only the problem statement and must produce its own reasoning and final answer. This point is important because the benchmark is not asking whether the model can reproduce a teacher rationale it has already seen. It asks whether the student can solve the problem independently after training.

The evaluation pipeline is intentionally kept fixed. In mathematical reasoning, reported scores may shift because of prompt wording, decoding parameters, output-length limits, or sampling choices. If those settings change across runs, it becomes much harder to judge whether an observed gain really comes from the reasoning data itself.That is why the same evaluation configuration is kept across the main comparison.

While developing the method, I also tracked several intermediate quantities, including answer recoverability when reasoning is given, step-level difficulty, and the local effect of candidate selection. These quantities are useful for understanding what the method is doing, but they are not the final thing I evaluate against. The main endpoint is still benchmark accuracy under problem-only inference, because that is the most direct way to judge whether the student model has really become a stronger solver.

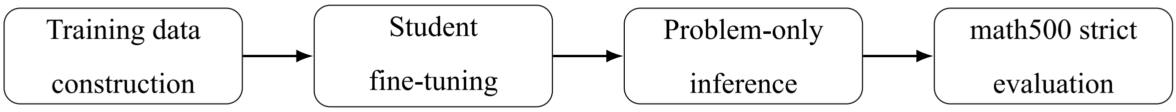

Figure 5 Evaluation protocol used in this thesis. The student model is first trained on the selected reasoning data, and its performance is then evaluated in a problem-only inference setting under the unified math500 strict protocol.

5 Experimental Results and Analysis

5.1 Overall Results

Table 2 Main Results on math500 strict

Model Scale	Untrained Base Model	Trained on Original CoTs	Trained on Fixed CoTs
Qwen2.5-Math-1.5B	27.4	70.0	73.8
Qwen2.5-Math-7B	48.8	76.8	78.8

Note: “Untrained Base Model” refers to the original model evaluated directly without fine-tuning. “Trained on Original CoTs” refers to the model fine-tuned on teacher-generated original CoT data without fix. “Trained on Fixed CoTs” refers to the model fine-tuned on the repaired CoT data after fix.

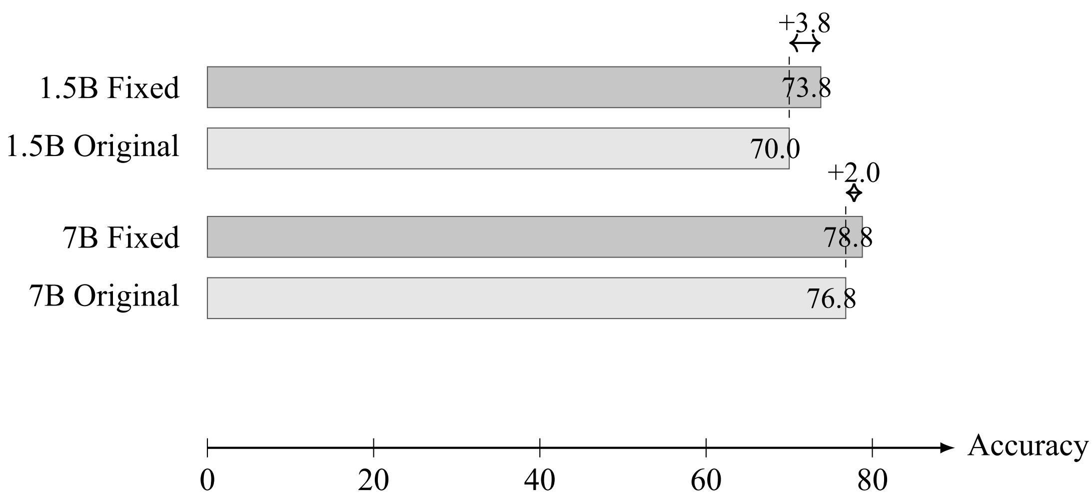

Figure 6 A visual comparison of student performance after training on original versus fixed CoTs. The gain is larger for the 1.5B student, while the 7B student still shows a clear improvement.

Table 2 supports several straightforward observations. First, both student scales benefit greatly from fine-tuning when compared with their untrained base versions. For the 1.5B model, the score rises from 27.4 to 70.0 after training on the original teacher-generated CoTs, and then further increases to 73.8 after training on the fixed CoTs. For the 7B model, the corresponding progression is 48.8 to 76.8 and then 78.8. This means that teacher-generated reasoning data already provides strong supervision by itself, and that repaired CoTs can still add extra value beyond the original training set.

Second, CoT fixing helps at both model scales, but the gains are not equally large. The 1.5B student gets a 3.8-point improvement over training on the original CoTs, while the 7B student gains 2.0 points. So the direction of the result is the same in both cases, but the smaller model seems to benefit more. This fits the basic intuition of the thesis quite well: when a student model has less internal reasoning capacity, it is usually more vulnerable to missing local bridges in the training trace.

SRFT can serve as a useful point of reference here. The SRFT paper reports that its single-stage framework, which combines supervised training and reinforcement learning, achieves an average improvement of more than 4.7% over other demonstration-based methods under its own post-training and evaluation pipeline [3]. This is relevant to the present thesis because SRFT also studies mathematical reasoning and is built on a carefully filtered training line derived from OpenR1-Math-46k-8192 together with high-quality reasoning responses [3]. At the same time, this should not be understood as a direct ranking at the benchmark level. SRFT changes the optimization objective and the training procedure, whereas the present thesis intervenes earlier by changing the quality of the reasoning data before supervised fine-tuning. For this reason, the comparison here is better treated as a contextual reference than as a direct head-to-head comparison.

From this point of view, the two results are better read as complementary. SRFT suggests that reasoning performance can be improved by changing the objective itself through a combination of SFT and RL. What the present thesis shows, on the other hand, is that even without touching the training objective, measurable gains can still come from refining CoTs on the data side under a relatively simple supervised fine-tuning setup.

5.2 Why the Improvements Are Meaningful

The improvements in Table 2 should be read carefully. In mathematical reasoning, benchmark scores may vary because of prompting details, sampling choices, or ordinary evaluation noise. For that reason, this thesis uses a unified math500 strict protocol and reports the main comparison under the same evaluation setting. Under this condition, a gain of 3.8 points on the 1.5B model and 2.0 points on the 7B model is already meaningful, especially because the proposed method does not change the student architecture and does not rely on a different evaluation procedure.

These gains also matter for another reason: they come from data optimization rather than model scaling or a more complex test-time setup.

5.3 Why Local Repair Helps

The results suggest a fairly simple but still important point. Even when teacher-generated CoTs all arrive at the correct final answer, they are not equally useful as supervision for student models. In practice, a CoT can still be a poor learning signal if some parts of it jump too quickly, compress too much derivation into one place, or leave the connection between key steps too implicit.

The method seems to help because it does not start from the assumption that every part of a CoT needs the same kind of intervention. What it tries to do instead is to find the steps that are both important and genuinely hard for the student, and then make a limited local repair there. This makes the final CoT easier to absorb without having to rewrite the whole solution path. So the gain appears to come not from making the reasoning longer in a general sense, but from making the critical local part easier to follow. If the edit scope is expanded without restraint, the rationale may look more complete on the surface, but the extra text is not always useful for the student and may even add new noise.

5.4 A Qualitative Example of Local Repair

A common step-level example makes it easier to see why local repair can matter. In an original teacher-generated CoT, one mathematically valid step may squeeze several intermediate relations into a single sentence. The reasoning may jump straight from the symbolic setup to a transformed expression, without writing out the intermediate substitution or algebraic transition in between. For a stronger teacher model, that level of compression may be perfectly manageable. But for a smaller student model, the same place can become exactly where the reasoning starts to break.

After repair, that same region is usually rewritten into a form that the student can follow more easily. Rather than changing the whole solution path, the repaired CoT may insert a short bridge step, break one dense transformation into two smaller steps, or state the role of an intermediate variable more clearly. The key point is that the repaired version is not simply longer for the sake of being longer. What changes is that the reasoning becomes more explicit exactly where the original version moved too fast.

This qualitative pattern also matches the basic design of the method. The gain does not appear to come mainly from turning the whole solution into a longer or more wordy explanation. What seems to matter more is finding the step that is both important and hard for the student, and then making that local region easier to learn from. This also offers a possible explanation for why the method is more effective on the 1.5B model than on the 7B model: the smaller student is simply more sensitive to missing local transitions.

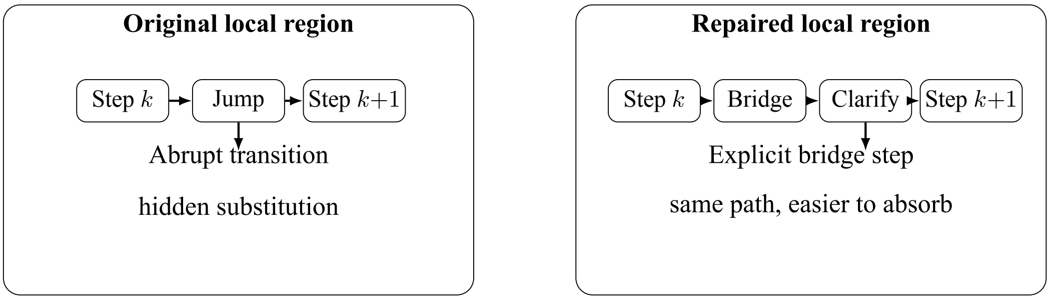

Figure 7 A schematic comparison of local repair. The original CoT may contain a compressed transition, whereas the repaired version inserts a bridge step or clarifies the dense transformation so that the same reasoning path becomes more learnable for the student model.

Table 3 Typical Effects of Local Repair at the Step Level

Repair Type	Typical Problem in Original CoT	Effect After Repair
Bridge Insertion	Abrupt jump between two symbolic steps	Makes the missing transition explicit
Step Split	One dense step contains multiple operations	Reduces the local reasoning burden per step
Local Clarification	Variable or quantity relation is under-specified	Makes the functional role of the step clearer
Answer Anchoring	Final conclusion is weakly connected to the derivation	Improves answer recoverability at the end

Table 3 does not report benchmark scores by itself, but it helps clarify what the repair mechanism is actually changing in practice. In that sense, it complements the main quantitative results by showing that the method mainly targets local reasoning form rather than replacing the whole solution globally.

5.5 Differences Across Model Scales

A clear pattern in Table 2 is that the 1.5B and 7B student models do not benefit in exactly the same way. The smaller model gains more in absolute terms, moving from 70.0 to 73.8, while the larger model improves from 76.8 to 78.8. A plausible reason is that the smaller student is more sensitive to local gaps in the reasoning. If a bridge step is missing or several key quantities are packed into one dense step, the 1.5B model is more likely to have trouble learning from that solution. From this perspective, repairing a small number of important local regions can clearly make the training data more useful.

For the larger model, the improvement is still visible, but the margin is smaller. A reasonable explanation is that the 7B model already has stronger internal reasoning and representation ability, so it depends less on local repair than the 1.5B model. Even so, the fact that the 7B model still gains 2.0 accuracy points suggests that the method remains useful even for a stronger student.

5.6 Why More Complexity Does Not Always Help

The experiments also suggest that increasing method complexity does not automatically improve the final result. It is easy to assume that more edit types, more candidate traces, or more elaborate scoring rules should always help. In practice, that expectation was not reliably supported. Some heavier variants looked more flexible on paper, but their downstream gains were often small or unstable.

A likely reason is that this task is sensitive to intervention scope. The main challenge is not to build the most complicated editing system, but to change the specific place that actually blocks student learning. Once the method becomes too aggressive, it may start rewriting regions that were not the real bottleneck in the first place. The output can look fuller and more polished while becoming less faithful to the original reasoning structure. From this point of view, careful and controlled repair matters more than simply making the method more complex.

5.7 Why Better Learnability Does Not Guarantee Higher Accuracy

A second important observation is that making reasoning easier to learn from does not automatically mean that standalone accuracy will rise by the same proportion. This is in line with recent work on CoT distillation, which shows that the link between supervision quality and final student performance is not a simple monotonic one, especially for smaller models [7, 8]. A trace may look cleaner, longer, or more explicit, but that still does not guarantee the strongest transfer to the student model.

One possible explanation is that two kinds of improvement are being mixed together. The first is improvement in imitation: the student finds the repaired trace easier to follow, predict, or reproduce. The second is improvement in independent solving: the student becomes better at deriving the answer when the teacher trace is no longer provided. These two outcomes are related, but they are not guaranteed to rise together. A revised CoT can reduce local expression difficulty without fully strengthening the strategic part of the solution that matters most at inference time.

This is why the contribution of the thesis should be read carefully. The point is not that repaired CoTs are universally superior in every sense. Rather, the experiments make clearer that correctness, clarity, and learnability describe different aspects of reasoning data. They often move in the same direction, but they do not collapse into one single notion of quality.

5.8 Limitations

This work still has a number of limitations. To begin with, although key-step localization uses multi-dimensional signals, it is still an approximate method based on student feedback rather than a genuine causal analysis of which step matters most. Some steps may look difficult at the local level even though they are not really decisive for the whole problem, while some truly decisive steps may not appear especially difficult when viewed on the surface. Because of this, there is still a gap between repairing the step that looks hardest and strengthening the step that is actually most critical, and that gap remains unresolved.

Second, although the method uses multi-stage candidate selection to preserve structure, local editing still depends on the quality and stability of Teacher-generated candidates. If the Teacher produces unstable local edits, the filtering mechanism can block obviously poor samples, but it cannot guarantee that the best possible candidate is always found.

Finally, the experiments still use unified benchmark accuracy as the main endpoint. More detailed quantitative analysis of convergence speed, stability under different random seeds, and in-domain reasoning-conditioned usefulness remains limited. If these aspects are measured more systematically in future work, it may become clearer what capability is actually being improved by the proposed method.

6 Conclusion and Future Work

This thesis studies a practical problem that became clear during the experiments: a teacher-generated Chain-of-Thought can be sufficient for solving the original problem and still be unsatisfactory as training data for a smaller student model. In many samples, the weakness does not lie in the whole rationale. More often, it appears in one narrow place where too much information has been compressed, such as a skipped bridge, a dense symbolic move, or a quantity relation that never gets written out. The method proposed here responds to that kind of case by locating the region that looks hardest from the student side and repairing only that part, rather than rewriting the full trace by default.

Under the experimental setting used in this thesis, this strategy performs better than directly training on the original traces. At the same time, the results also show a clear limit. A rationale can become easier to imitate, easier to read, or more stable as supervision without producing the same scale of gain in independent benchmark solving. For that reason, it is not enough to describe reasoning data with a single label such as good or bad. Correctness, clarity, and learnability may move together in some cases, but they should not be treated as identical.

So the contribution of this thesis is limited, but it is still quite concrete. On the methodological side, it offers a workable way to look at teacher CoTs below the level of the whole sample and revise them while keeping the student’s learning burden in view. On the empirical side, it suggests that local reasoning form is not merely a matter of presentation. Even if the problem and the final answer stay the same, changing one compact but important part of the reasoning can still make a real difference to how useful the sample is for training.

There are still several directions worth exploring in future work. One is to go beyond simply finding the locally hardest step and instead ask which local step is actually most decisive for final solving success. Another is to combine this kind of data-side intervention with stronger optimization objectives and more systematic filtering. It would also be useful to look more carefully at where the observed gain is really coming from, for example whether it mainly shows up in easier imitation, changes in convergence behavior, or genuinely stronger independent reasoning. Overall, the current results suggest that student-oriented CoT revision is worth studying further, but it is probably better viewed as one part of a broader reasoning post-training pipeline rather than a complete solution by itself.

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